PES UNIVERSITY

100-ft Ring Road, BSK III Stage, Bangalore – 560 085

Department of Computer Science & Engineering

Session: AUG-DEC 2024

UE23MA242A: Mathematics for Computer Science Engineers

Classwork Problems

#	UNIT_3 Solutions	Exercise Number
1	In an experiment to measure the lifetimes of parts manufactured from a certain aluminium alloy, 73 parts were loaded cyclically until failure. The mean number of kilocycles to failure was 783, and the standard deviation was 120. Let μ represent the mean number of kilocycles to failure for parts of this type. A test is made of H0: $\mu \le 750$ versus H1: $\mu > 750$. a) Find the P-value. b) Either the mean number of kilocycles to failure is greater than 750, or the sample is in the most extreme__________ % of its distribution. Ans: (a) $\bar{x}$ =783, s=120, n=73. The null and alternate hypotheses are H0: $\mu \le 750$ versus H1: $\mu > 750$. $z = (783-750)/(120/\sqrt{73}) = 2.35$. Since the alternate hypothesis is of the form $\mu > \mu_0$, the P-value is the area to the right of $z = 2.35$. Thus P=0.0094. (b) The P-value is 0.0094, so if H0 is true then the sample is in the most extreme 0.94% of its distribution.	**Section 6.1 Q. No: 1**
2	A certain type of stainless steel powder is supposed to have a mean particle diameter of μ =15μm. A random sample of 87 particles had a mean diameter of 15.2 μm, with a standard deviation of 1.8 μm. A test is made of H0: $\mu = 15$ versus H1: $\mu = 15$. a) Find the P-value. b) Do you believe it is plausible that the mean diameter is 15 μm, or are you convinced that it differs from 15 μm? Explain your reasoning. Ans: (a)$\bar{x}$=15.2, s =1.8, n =87. The null and alternate hypotheses are H0: $\mu = 15$ versus H1: $\mu = 15$. $z = (15.2-15)/(1.8/\sqrt{87}) = 1.04$.Since the alternate hypothesis is of the form $\mu = \mu_0$, the P-value is the sum of the areas to the right of $z = 1.04$ and to the left of $z = -1.04$. Thus P=0.1492+0.1492=0.2984.	**Section 6.1 Q. No: 6**

	(b) If the mean particle diameter were 15 µm, the probability of observing a sample mean as far from 15 as the value of 15.2 that was actually observed would be 0.2984. Since 0.2984is not a small probability, it is plausible that the mean is 15 µm.	
3	For which P-value is the null hypothesis more plausible: P = 0.5 or P = 0.05? Ans: P=0.5. The larger the P-value, the more plausible the null hypothesis.	**Section 6.2 Q. No: 1**
4	**True or false:** a) If we reject H0, then we conclude that H0 is false. b) If we do not reject H0, then we conclude that H0 is true. c) If we reject H0, then we conclude that H1 is true. d) If we do not reject H0, then we conclude that H1 is false. Ans: (a) True. (b) False. We conclude that H0 is plausible. (c) True. (d) False. We conclude that H0 is plausible, and therefore that H1 is plausible as well.	**Section 6.2 Q. No: 2**
5	If P = 0.01, which is the best conclusion? i. H0 is definitely false. ii. H0 is definitely true. iii. There is a 1% probability that H0 is true. iv. H0 might be true, but it's unlikely. v. H0 might be false, but it's unlikely. vi. H0 is plausible. Ans: (iv). A P-value of 0.01 means that if H0 is true, then the observed value of the test statistic was in the most extreme 1% of its distribution. This is unlikely, but not impossible.	**Section 6.2 Q. No: 3**
6	If P = 0.50, which is the best conclusion? i. H0 is definitely false. ii. H0 is definitely true. iii. There is a 50% probability that H0 is true. iv. H0 is plausible, and H1 is false. v. Both H0 and H1 are plausible. Ans: (v). A P-value of 0.50 means that if H0 is true, then the observed value of the test statistic was in the most extreme 50% of its distribution. This is not at all unusual, so H0 is plausible. We can never conclude that H1 is false, so therefore we conclude that H1 is plausible as well.	**Section 6.2 Q. No: 4**

7	Let μ be the radiation level to which a radiation worker is exposed during the course of a year. The Environmental Protection Agency has set the maximum safe level of exposure at 5 rems per year. If a hypothesis test is to be performed to determine whether a work place is safe, which is the most appropriate null hypothesis: H0: $\mu \leq 5$, H0: $\mu \geq 5$, or H0: $\mu = 5$? Explain. Ans: H0: $\mu \geq 5$is the best. If H0 is rejected, we can be convinced that $\mu < 5$, and that the workplace is safe. If H0 is not rejected, then the workplace might not be safe.	**Section 6.2** **Q. No: 8**
8	A random sample of 300 electronic components manufactured by a certain process are tested, and 25 are found to be defective. Let p represent the proportion of components manufactured by this process that are defective. The process engineer claims that $p \leq 0.05$. Does the sample provide enough evidence to reject the claim? Ans: X =25, n=300, $\hat{p}$=25/300=0.083333. The null and alternate hypotheses are H0: $p \leq 0.05$ versus H1: $p > 0.05$. z =(0.083333−0.5)/ √(0.05(1−0.05)/300)= 2.65. Since the alternate hypothesis is of the form p > p0, the P-value is the area to the right of z = 2.65, so P=0.0040. The claim is rejected.	**Section 6.3** **Q. No: 1**
9	In a survey of 500 residents in a certain town, 274 said they were opposed to constructing a new shopping mall. Can you conclude that more than half of the residents in this town are opposed to constructing a new shopping mall? Ans: X =274,n=500, $\hat{p}$=274/500=0.548. The null and alternate hypotheses are H0: $p \leq 0.50$ versus H1: $p > 0.50$. z =(0.548−0.50)/ √(0.50(1−0.50)/500)=2.15. Since the alternate hypothesis is of the form p > p0, the P-value is the area to the right of z = 2.15, so P=0.0158. We can conclude that more than half of residents are opposed to building a new shopping mall.	**Section 6.3** **Q. No: 5**
10	(a) $H_0: \mu \leq 5$ vs. $H_1: \mu > 5$ (b) $t_7 = 2.2330$, $0.025 < P < 0.05$ ($P = 0.03035$). (c) Yes, the P-value is small, so we conclude that the mean flow rate is more than 5 gpm.	**Section 6.4** **Q. No: 3**

11.

(a) The signed ranks are

x	$x-41$	Signed Rank
41.01	0.01	1
41.37	0.37	2
41.42	0.42	3
40.50	−0.50	−4
41.70	0.70	5
41.83	0.83	6.5
41.83	0.83	6.5
42.68	1.68	8

The null and alternate hypotheses are $H0:\mu\le 41$ versus $H1:\mu>41$. The sum of the positive signed ranks is $S+=32$. $n=8$. Since the alternate hypothesis is of the form $\mu>\mu0$, the P-value is the area under the signed rank probability mass function corresponding to $S+\ge 32$. From the signed-rank table, $P=0.0273$. We can conclude that the mean thickness is greater than 41mm.

Section 6.9
Q. No: 2

(b) The signed ranks are

x	$x-41.8$	Signed Rank
41.83	0.03	1.5
41.83	0.03	1.5
41.70	−0.10	−3
41.42	−0.38	−4
41.37	−0.43	−5
41.01	−0.79	−6
42.68	0.88	7
40.50	−1.30	−8

The null and alternate hypotheses are $H0:\mu\ge 41.8$ versus $H1:\mu<41.8$.
The sum of the positive signed ranks is $S+=10$. $n=8$.
Since the alternate hypothesis is of the form $\mu<\mu0$, the P-value is the area under the signed rank probability mass function corresponding to $S+\le 10$.
From the signed-rank table, $P>0.1250$.
We cannot conclude that the mean thickness is less than 41.8mm.

(c) The signed ranks are

x	$x-42$	Signed Rank
41.83	−0.17	−1.5
41.83	−0.17	−1.5
41.70	−0.30	−3
41.42	−0.58	−4
41.37	−0.63	−5
42.68	0.68	6
41.01	−0.99	−7
40.50	−1.50	−8

The null and alternate hypotheses are $H0:\mu=42$ versus $H1:\mu\neq 42$.
The sum of the positive signed ranks is $S+=6$. $n=8$.
Since the alternate hypothesis is of the form $\mu\neq\mu0$, the P-value is the area under the signed rank probability mass function corresponding to $S+\le 6$.
From the signed-rank table, $P=2(0.0547)=0.1094$.
We cannot conclude that the mean thickness differs from 42mm.

12.

Ranks of combined samples are:

Value	Rank	Sample
0.43	1	Y
0.73	2	X
0.93	3	Y
1.12	4	X
1.24	5	X
1.91	6	Y
2.56	7	Y
2.93	8	X
3.72	9	Y
6.19	10	Y
11.00	11	Y

The null and alternate hypotheses are $H0:\mu X - \mu Y = 0$ versus $H1:\mu X - \mu Y = 0$.
The test statistic W is the sum of the ranks corresponding to the X sample. $W=19$.
The sample sizes are $m=4$ and $n=7$. Since the alternate hypothesis is of the form $\mu X - \mu Y = \Delta$, the P-value is twice the area under the rank-sum probability mass function corresponding to $W \leq 19$.
From the rank-sum table, $P>2(0.0545) = 0.1090$.
We cannot conclude that the mean rates differ.

Section 6.9
Q. No: 6

13.

Ranks of combined samples are:

Value	Rank	Sample
51	1	Y
59	2	Y
64	4	X
64	4	Y
64	4	Y
66	6	X
69	7	X
72	8	X
74	9.5	X
74	9.5	Y
75	11.5	X
75	11.5	Y
77	14	X
77	14	X
77	14	Y
80	16.5	Y
80	16.5	Y
81	18.5	Y
81	18.5	Y
83	20.5	X
83	20.5	Y
85	22.5	X
85	22.5	Y
86	24	Y
88	25	X
91	26	X

The null and alternate hypotheses are $H0:\mu X - \mu Y = 0$ versus $H1:\mu X - \mu Y \neq 0$.
The test statistic W is the sum of the ranks corresponding to the X sample. $W=168$.
The sample sizes are $m=12$ and $n=14$.
Since n and m are both greater than 8, compute the z-score of W and use the z table.

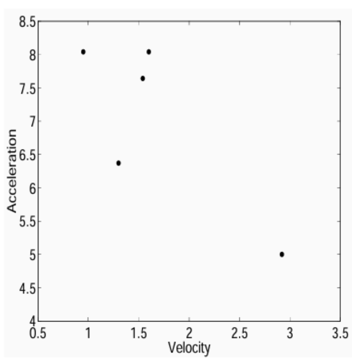
$$z = \frac{W - m(m+n+1)/2}{\sqrt{mn(m+n+1)/12}} = 0.31.$$

Since the alternate hypothesis is of the form $\mu X - \mu Y = \Delta$, the P-value is the sum of the areas to the right of $z=0.31$ and to the left of $z=-0.31$. From the z table, $P=0.3783+0.3783=0.7566$.
We cannot conclude that the mean scores differ.

Section 6.9
Q. No: 9

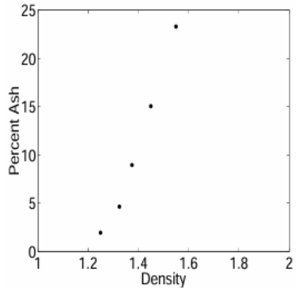
14.	(a) Let pij denote the probability that an observation is in column j, given that it is in row i. Then the null hypothesis is H0: For each column j, p1j = p2j = p3j. (b)The row totals are O1.=700,O2.=650,O3.=400. The column totals are O.1=1166,O.2=491,O.3=93. The grand total is O..=1750. The expected values are Eij=Oi.O.j/O.., as shown in the following table. 466.400 196.400 37.200 433.086 182.371 34.543 266.514 112.229 21.257 (c) $\chi^2 = \sum_{i=1}^{3}\sum_{j=1}^{3}(O_{ij} - E_{ij})^2/E_{ij} = 5.760$ (d)There are (3−1)(3−1)=4 degrees of freedom. From the χ2 table, P>0.10. A computer package gives P=0.218. There is no evidence that the quality of welds differs among the shifts.	**Section 6.10** **Q. No: 2**
15.	(a) The row totals are O1.=37, O2.=25, and O3.=35. The column totals are O.1=27, O.2=35, O.3=35. The grand total is O..= 97. The expected values are Eij=Oi.O.j/O.., as shown in the following table. 10.2990 13.3505 13.3505 6.9588 9.0206 9.0206 9.7423 12.6289 12.6289 (b) The χ2 test is appropriate here, since all the expected values are greater than 5. There are (3−1)(3−1) = 4 degrees of freedom. $\chi^2 = \sum_{i=1}^{3}\sum_{j=1}^{3}(O_{ij} - E_{ij})^2/E_{ij} = 6.4808.$ From the χ2 table, P>0.10. A computer package gives P=0.166. There is no evidence that the rows and columns are not independent.	**Section 6.10** **Q. No: 7**

16.	(a) Let $\underline{x}$ be the sample mean of the 80 specimens. Under H0, the population mean is μ = 2, and the population standard deviation is σ = 0.6. The null distribution of $\underline{x}$ is therefore normal with mean 2 and standard deviation 0.6/√80 = 0.0670820. Since the alternate hypothesis is of the form μ< μ0, the rejection region for a 5% level test consists of the lower 5%of the null distribution. The z-score corresponding to the lower 5% of the normal distribution is z = −1.645. Therefore the rejection region consists of all values of $\underline{x}$ less than or equal to 2−1.645(0.0670820)= 1.890. H0 will be rejected if $\underline{x}$ ≤ 1.890. (b) Since the alternate hypothesis is of the form μ < μ0, the rejection region for a 10% level test consists of the lower 10% of the null distribution. The z-score corresponding to the lower 10% of the normal distribution is z = −1.28. Therefore the rejection region consists of all values of $\underline{x}$ less than or equal to 2−1.28(0.0670820)= 1.9141. Since 1.85 < 1.9141, H0 will be rejected at the 10% level. (c) Since the alternate hypothesis is of the form μ< μ0, the rejection region for a 1% level test consists of the lower 1%of the null distribution. The z-score corresponding to the lower 1% of the normal distribution is z = −2.33. Therefore the rejection region consists of all values of X less than or equal to 2−2.33(0.0670820)= 1.8437. Since 1.85 > 1.8437, H0 will not be rejected at the 1% level. (d) Under H0, the z-score of 1.9 is (1.9−2)/0.0670820= −1.49. Since the alternate hypothesis is of the form μ < μ0, the level is the area to the left of z = −1.49. Therefore the level is α = 0.0681.	**Section 6.12** **Q. No: 2**
17.	(a) Type I error. H0 is true and was rejected. (b) Correct decision. H0 is true and was not rejected. (c) Type II error. H0 is false and was not rejected. (d) Correct decision. H0 is false and was rejected.	**Section 6.12** **Q. No: 4**
18.	The maximum probability of rejecting H0 when true is the level α = 0.10.	**Section 6.12** **Q. No: 6**
19.	(a) True. This is the definition of power. (b) True. When H0 is false, making a correct decision means rejecting H0. (c) False. The power is 0.90, not 0.10. (d) False. The power is not the probability that H0 is true.	**Section 6.13** **Q. No: 1**
20.	$\overline{x}=\overline{y}=4,\ \ \sum_{i=1}^{n}(x_i-\overline{x})^2=28,\ \ \sum_{i=1}^{n}(y_i-\overline{y})^2=28,\ \ \sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y})=$ $r=\dfrac{\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\overline{y})^2}}=0.8214.$	**Section 7.1** **Q. No: 1**

21.	(a) y has been replaced with y+3, x is unchanged. This involves only adding a constant, which does not change the correlation coefficient. (b) x has been replaced with 10x+1, y has been replaced with y+3. This involves only adding a constant and multiplying by a constant, neither of which changes the correlation coefficient. (c) x has been replaced with 10y+33, y has been replaced with 2x+2. This involves only adding a constant, multiplying by a constant, and interchanging x and y, none of which changes the correlation coefficient.	**Section 7.1** **Q. No: 2**
22.	(a) True. This is a result of the plot being clustered around a line with positive slope. (b) False. Since the plot is clustered around a line with negative slope, below average values of one variable will tend to be associated with above average values of the other. (c) False. The correlation coefficient describes the shape of the plot, not its location relative to the axes.	**Section 7.1** **Q. No: 4**
23.	(a) Let x represent velocity and let y represent acceleration. $\overline{x} = 1.662$, $\overline{y} = 7.018$, $\sum_{i=1}^{n}(x_i - \overline{x})^2 = 2.23928$, $\sum_{i=1}^{n}(y_i - \overline{y})^2 = 6.96808$, $\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y}) = -3.17098$. $r = \dfrac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}} = -0.8028.$ (b)  (c) No, the point (2.92, 5.00) is an outlier. (d) No effect. Converting units from meters to centimetres and from seconds to minutes involves multiplying by a constant, which does not change the correlation coefficient.	**Section 7.1** **Q. No: 6**
24.	(a) 245.82+1.13(65)= 319.27 pounds (b) The difference in y predicted from a one-unit change in x is the slope β1 = 1.13. Therefore the change in the number of lbs of steam predicted from a change of 5◦C is 1.13(5)= 5.65 pounds.	**Section 7.2** **Q. No: 1**
25.	$r^2 = 1 - \dfrac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2} = 1 - \dfrac{33.9}{181.2} = 0.8129.$	**Section 7.2** **Q. No: 4**

26.

(a)



The linear model is appropriate.

Section 7.2
Q. No: 10

(b) $\overline{x} = 1.39$, $\overline{y} = 10.774$, $\sum_{i=1}^{n}(x_i - \overline{x})^2 = 0.05325$, $\sum_{i=1}^{n}(y_i - \overline{y})^2 = 294.72752$,

$\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y}) = 3.9272$.

$\widehat{\beta}_1 = \dfrac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = 73.75023$ and $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x} = -91.7388$.

The equation of the least-squares line is $y = -91.7388 + 73.75023x$.

(c) 73.75023(0.1)= 7.375%

(d) −91.7388+73.75023(1.40)= 11.51%

(e) The fitted values are the values $y_i = \beta_0+\beta_1 x_i$, for each value x_i. They are shown in the following table.

x	y	Fitted Value $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$
1.25	1.93	0.449
1.325	4.63	5.980
1.375	8.95	9.668
1.45	15.05	15.199
1.55	23.31	22.574

(f) The residuals are the values $e_i = y_i - y_i$ for each i. They are shown in the following table.

x	y	Fitted Value $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$	Residual $e = y - \widehat{y}$
1.25	1.93	0.449	1.481
1.325	4.63	5.980	−1.350
1.375	8.95	9.668	−0.718
1.45	15.05	15.199	−0.149
1.55	23.31	22.574	0.736

The point whose residual has the largest magnitude is (1.25, 1.93).

(g) $r = \dfrac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}} = 0.991318.$

(h) The error sum of squares is $\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = (1 - r^2)\sum_{i=1}^{n}(y_i - \overline{y})^2 = 5.10$.

The regression sum of squares is $\sum_{i=1}^{n}(y_i - \overline{y})^2 - \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = 289.63$.

The total sum of squares is $\sum_{i=1}^{n}(y_i - \overline{y})^2 = 294.73$.

	(i) The quotient of the regression sum of squares divided by the total sum of squares is 289.63/294.73= 0.9827. This is the square of the correlation coefficient. To see this, note that $\frac{\sum_{i=1}^{n}(y_i-\overline{y})^2-\sum_{i=1}^{n}(y_i-\widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\overline{y})^2}=1-\frac{\sum_{i=1}^{n}(y_i-\widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\overline{y})^2}=1-(1-r^2)=r^2.$	